

Day 1 Schedule — Monday, October 5, 2026

"Landing your data in BigQuery" · 9:00 AM–4:30 PM CT · virtual. Day 1 is about console confidence and getting data in: loading files, streaming database changes live, and your first Dataform pipeline.

Time	Block	What you'll walk away with
09:00–09:25	Welcome & your migration story	The shape of the two days: kill the extracts → land in BigQuery → Bronze/Silver/Gold → your existing Power BI reports
09:25–10:10	DE building blocks on GCP	The GCP moving parts — projects, datasets, Cloud Storage, BigQuery — mapped to the sources/transforms/sinks model you already think in
10:10–10:25	Hands-on: find your way around the console	Three starred projects (bigquery-samples, bigquery-public-data, roi-bq-demos) and your own class dataset, created by you (DIN §3 — see note below)
10:25–10:40	Break	
10:40–11:10	Lab: your first BigQuery load — <i>Loading Data into BigQuery</i>	Your first finished lab in an unfamiliar console: a real table, loaded by you
11:10–11:50	Killing the nightly extracts	Why log-based change data capture (Datastream) replaces the extract window — SQL Server is a supported source, with no triggers and no load on production
11:50–12:00	Lab: set up a live	A running stream

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	replication stream (part 1) — <i>Datastream: PostgreSQL → BigQuery</i>	replicating a database into BigQuery before lunch
12:00–13:00	Lunch — your stream keeps replicating	
13:00–13:15	Lab part 2: watch changes land	The CDC moment, felt: rows changed at the source appear in BigQuery with no job run
13:15–13:45	Loading files the right way	The extract-and-load pattern for file drops: console loads, the bq CLI, scheduled ingest with the BigQuery Data Transfer Service, and when an external table beats a load
13:45–14:30	Hands-on: four ways to load (and the format race)	Console & CLI load, the autodetect N/A pitfall, the 60-million-row CSV-vs-Parquet race, and LOAD DATA SQL (DIN §24–§27) — plus the rule: pin schemas for anything recurring
14:30–14:45	Break	
14:45–15:25	ELT & Dataform — your Silver→Gold tool	How Dataform turns your SQL into a versioned, tested, scheduled pipeline — the tool your own architecture doc already names
15:25–16:05	Lab: build a SQL workflow in Dataform — <i>Create & Execute a SQL Workflow in Dataform</i>	A working workflow you built yourself: models, dependencies, assertions
16:05–16:20	When SQL isn't enough + automation	The escape hatches (Pub/Sub, Dataflow) and the automation map:

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		Cloud Scheduler ≈ SQL Agent, Composer ≈ the orchestrator, Dataform release configs ≈ your CI/CD
16:20–16:30	Wrap-up: schedule your first query	Your first piece of BigQuery automation: a query on a daily schedule overwriting a table (DIN §29, step 5) — no SQL Agent box required

Worth knowing before you arrive

- **Every block title above links to its module notes or lab/activity companion in this folder.** Open the companions before the hands-on blocks — they carry the gotchas.
- **The class dataset matters.** You create it yourself at 10:10, in your own lab project. Every loading activity this afternoon writes into it — it's the one step that can't be skipped or made up later.
- **Labs live in the [Google Skills classroom](#).** Launching a lab starts its timer and hands you a temporary GCP project. The lab's own steps are in the classroom; the companions in this folder add orientation and gotchas.
- **'DIN' = the Do-It-Now activities page:**
<https://roittraining.github.io/gcp-demos/bigquery.html> — a public set of short BigQuery exercises referenced by section number (§) all day. All four loading DINs this afternoon are free.
- **The Datastream lab uses PostgreSQL** because it's free-tier friendly. The mechanics — connection, stream, backfill, change capture — are identical for SQL Server, which is a supported Datastream source ([docs](#)).
- **Demo data you'll touch today:** the public roi-bq-demos and bigquery-public-data projects, and the public GCS bucket jwd-gcp-demos.

Reference cards to keep open today

- [BigQuery console survival guide](#) — during the 10:10 console orientation and every lab
- [T-SQL → BigQuery SQL](#) — the moment you start typing queries
- [GCP for SQL Server professionals](#) — the mapping card behind every block today

Part of the Iowa DOM DoIT Data Engineering student pack — see ../README.md.